



LINEAR PATH AS AN EXPONENTIALLY

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η is the base-measure

► β is the natural parameter where



$V \equiv \log v$ is the sufficient statistic

► The linear path extends naturally to an exponential family $\mathcal{E} = \{\pi_\beta : \beta \in \Omega\}$

$$\pi_{\beta}(x) \equiv \exp(\beta V(x) - A(\beta)) \eta(x)$$



$\Omega = \{\beta : Z(\beta) < \infty\}$ is an interval containing $[0, 1]$

► $A(\beta) = \log Z(\beta)$ is the cumulant generating function

► In particular, the smoothness of $A(\beta)$ controls the moments of V at π_β

$$A'(\beta) = \pi_{\beta}[V]$$

► The second derivative is the variance of V with respect to π_β i.e. the Fisher information

$$A'(\beta) = \sqrt{\beta[V]} = I(\beta)$$

► The first derivative is the expectation of V with respect to π_β



Hence $\beta \mapsto A(\beta)$ is convex with $A(0) = 0$

LINEAR PATH AS AN EXPONENTIAL FAMILY

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- ▶ The linear path extends naturally to an exponential family $\mathcal{E} = \{\pi_\beta : \beta \in \Omega\}$

$$\pi_\beta(x) = \exp(\beta V(x) - A(\beta))\eta(x)$$

- ▶ $\Omega = \{\beta : Z(\beta) < \infty\}$ is an interval containing $[0,1]$
- ▶ η is the base-measure
- ▶ β is the natural parameter where
- ▶ $V = \log w$ is the sufficient statistic
- ▶ $A(\beta) = \log Z(\beta)$ is the cumulant generating function
- ▶ In particular, the smoothness of $A(\beta)$ controls the moments of V at π_β
 - ▶ The first derivative is the expectation of V with respect to π_β
- ▶ The second derivative is the variance of V with respect to π_β i.e. the Fisher information

$$A'(\beta) = \pi_\beta[V]$$
$$A''(\beta) = \mathbb{V}_\beta[V] = I(\beta)$$

- ▶ Hence $\beta \mapsto A(\beta)$ is convex with $A(0) = 0$

ANNEALING AS ROBUST BAYESIAN INFERENCE